

Tomás Pelayo-Benedet

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PROFESSIONAL EXPERIENCE

- **University of Zaragoza** Zaragoza, Spain
Ph.D. Researcher May 2024 – Present
 - Research on Algorithmically Generated Domains (AGD) detection using LLMs.
 - Design and benchmark deep learning architectures for AGD detection.
 - Review and benchmark adversarial AGD generation and defense techniques.
 - Research on the application of agents to Reverse Engineering.
- **Lecturer** Oct 2025 – May 2026
 - Teaching practical sessions for undergraduate engineering courses.
- **JnT Studios** Spain
Co-Founder Dec 2025 – Present
 - Lead both the planning and hands-on implementation of AI product.
 - Contribute developing core software features and infrastructure.
- **Ruhr University Bochum** (*Chair for Software Security, Prof. Dr. Kevin Borgolte*) Bochum, Germany
Visiting Researcher Apr 2025 – Jul 2025
 - Research on novel applications of LLMs for binary decompilation via intermediate representations.
- **Teltronic** Zaragoza, Spain
Cybersecurity Engineer May 2023 – May 2024
 - Lead cybersecurity strategy and compliance for LTE (4G) product line.
 - Implement risk mitigation policies aligned with industry standards.
- **Cybersecurity Intern** Oct 2022 – Apr 2023
 - Conduct vulnerability assessments and risk analysis for telecommunications systems.

EDUCATION

- **Ph.D. in Computer Science** 2024 – Present
University of Zaragoza
Research focus: AI applications in cybersecurity
- **Master of Business Administration (MBA)** 2026 – Present
International University of La Rioja
Expected graduation: 2027
- **Master in Research in Cybersecurity** 2023
University of León
Thesis: Detection of Algorithmically Generated Domains (AGDs)
- **Bachelor in Computer Science** 2022
University of Zaragoza
Graduated with honors in Artificial Intelligence

CERTIFICATIONS, COURSES & OTHER OFFICIAL STUDIES

- **University Expert in Cyber Incident Management and Response** – University of Zaragoza 2025
- **Containment and Forensics Expert after a Cyber Incident** – INCIBE & EICYC 2023
- **B2 Cambridge English: First (FCE)** – Cambridge University 2023
- **Elementary and Professional Music Studies** – CPMZ 2011 – 2022

SKILLS

- **Programming:** Python, C/C++, Java, Rust, Go, JavaScript, SQL, R, Bash, Assembly
- **Machine Learning:** Deep Learning, NLP, Large Language Models, TensorFlow, PyTorch
- **Cybersecurity:** Vulnerability Assessment, Risk Analysis, Incident Response, Malware Detection
- **Tools:** Git, Docker, Linux, LaTeX, Wireshark, Nmap, Burp Suite, SIEM platforms, Binary Ninja...
- **Cloud & Infrastructure:** AWS, Kubernetes, Jenkins
- **Languages:** Spanish (Native), English (B2+)

AWARDS, GRANTS & HONORS

- **Pre-doctoral Research Fellowship, Regional Government of Aragón (2025)**
(*Subvenciones para la contratación de personal investigador predoctoral en formación*)
- **I3A Bridge Research Grant (2025)**
(*Ayudas para la cofinanciación de Contratos Puente de Investigación del I3A*)

TALKS & CONFERENCES

- **Exploring the Zero-Shot Potential of Large Language Models for Detecting Algorithmically Generated Domains.** *Advanced Methods for Evidence-based Cybersecurity Research (EMACS) Starting Workshop.* Jaca, Aragón, España, 2026.
- **Does Representation Matter? Evaluating IRs for LLM-based Binary Decompilation.** *9th Workshop on Binary Analysis Research (BAR)*, co-located with NDSS. San Diego, CA, USA, 2026.
- **A review of “The Machines are Watching: Exploring the Potential of Large Language Models for Detecting Algorithmically Generated Domains”.** *XI Jornadas Nacionales de Investigación en Ciberseguridad (JNIC).* Barcelona, Cataluña, España, 2026.

PUBLICATIONS

- Pelayo-Benedet, T., & Rodríguez, R. J. (2026). A Systematic Literature Review of Adversarial Domain Generation and Defense. *Machine Learning with Applications*, 24, 100888. DOI: 10.1016/j.mlwa.2026.100888
- Pelayo-Benedet, T., Borgolte, K., & Rodríguez, R. J. (2026). Does Representation Matter? Evaluating IRs for LLM-based Binary Decompilation. In *Proceedings of the 9th Workshop on Binary Analysis Research (BAR)*. San Diego, CA, USA: Internet Society (ISOC). DOI: 10.14722/bar.2026.23077
- Pelayo-Benedet, T., Rodríguez, R. J., & Gañán, C. H. (2025). The machines are watching: Exploring the potential of Large Language Models for detecting Algorithmically Generated Domains. *Journal of Information Security and Applications*, 93, 104176. DOI: 10.1016/j.jisa.2025.104176
- Pelayo-Benedet, T., Rodríguez, R. J., & Gañán, C. H. (2025). Poster: Exploring the Zero-Shot Potential of Large Language Models for Detecting Algorithmically Generated Domains. In *International Conference on Detection of Intrusions and Malware, and Vulnerability Assessment* (pp. 86-92). Cham: Springer Nature Switzerland. DOI: 10.1007/978-3-031-97623-0_5
- Pelayo-Benedet, T., Rodríguez, R. J., & Gañán, C. H. (2025). RAMPAGE: A Software Framework To Ensure Reproducibility in Algorithmically Generated Domains Detection. *Expert Systems with Applications*, 128629. DOI: 10.1016/j.eswa.2025.128629
- Pelayo, T., Arronategui, U., Bañares, J. Á., & Colom, J. M. (2022). Design and Implementation of a Compiler for Simulation of Large-Scale Models. In *International Conference on the Economics of Grids, Clouds, Systems, and Services* (pp. 158-162). Cham: Springer Nature Switzerland. DOI: 10.1007/978-3-031-29315-3_16